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Monoidal categories from vertex operator algebras – beyond rigidity

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Joint work with
Jürgen Fuchs, Gregor Schaumann, Christoph Schweigert, arXiv:[2306.17668](#), [2405.20811](#)
Robert Allen, Simon Lentner, Christoph Schweigert arXiv:[2107.05718](#)

Recent Developments in Logarithmic Conformal Field Theory
Banff International Research Station 2025

Rigidity

Definition

- 1 An object X in a monoidal category $(\mathcal{C}, \otimes)$ admits a *rigid right dual* $X^\vee$, if there exist morphisms

$$\mathrm{ev}^r : X^\vee \otimes X \rightarrow \mathbf{1}, \quad \mathrm{coev}^r : \mathbf{1} \rightarrow X \otimes X^\vee,$$

satisfying zig-zag relations.

- 2 *Rigid left duals* are defined similarly.
- 3 An object $X \in \mathcal{C}$ is called *rigid* if it admits both a left and a right dual.

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Examples

- 1 Category of finite-dimensional modules over a finite-dimensional Hopf algebra.
- 2 Category of admissible modules over a rational c_2 -cofinite vertex operator algebra.

Consequences of rigidity

- 1 In abelian rigid monoidal categories monoidal products are **exact**.
- 2 Projective objects form a monoidal ideal.
- 3 Taking duals is opmonoidal: $(c \otimes d)^\vee \cong d^\vee \otimes c^\vee$, that is $(-)^\vee : (\mathcal{C}, \otimes) \rightarrow (\mathcal{C}^{\text{opp}}, \otimes^{\text{opp}})$.

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- 1 For A a finite-dimensional associative algebra and $B \in A\text{-Bimod}$. Then $B \otimes_A -$ is generally only **right exact**.
- 2 V a vertex operator algebra and $M \in V\text{-Mod}$ (to which HLZ theory applies). Then $M \otimes_V -$ is generally only **right exact** (see (p, q) -triplet $p, q \geq 2$, $\gcd(p, q) = 1$).

Grothendieck-Verdier categories / $*$ -autonomous categories

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A *Grothendieck-Verdier* (GV) category is a monoidal category $(\mathcal{C}, \otimes)$ together with a choice of dualising object $K \in \mathcal{C}$.

Basic properties and examples

Proposition [Boyarchenko, Drinfeld '13]

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Let V be a vertex operator algebra and $\mathcal{C}$ a category of V -modules (HLZ applies, $V \in \mathcal{C}$, closed under graded duals).

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Theorem [Allen '23], [Demirdelik '24]

Let H be a Hopf algebroid with finite-dimensional base algebra A and an invertible antipode S . Then the category of finite-dimensional H -modules is a GV category. A dualising object is given by the vector space dual of the base algebra A .

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Proposition

Every simple $\mathbb{C}_{\kappa} \in \text{Vec}_{\mathbb{C}}(\mathbb{R})$ is dualising with dualising functor $G(M) = \bigoplus_{\lambda} G(M)_{\lambda}$, $G(M)_{\lambda} = M_{\kappa-\lambda}$. There is a ribbon twist $\theta(\mathbb{C}_{\lambda}) = e^{i\pi\lambda(\lambda-\kappa)} \text{id}$.

Important consequences of GV structures

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Now consider $\mathcal{C}$ as a monoidal category with **two** monoidal products: $\otimes$ and $\boxtimes$.

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- ① $- \otimes Y$ admits a right adjoint (internal Hom):

$$\mathrm{Hom}(X \otimes Y, Z) \cong \mathrm{Hom}(X, \underline{\mathrm{Hom}}(Y, Z)), \text{ with } \underline{\mathrm{Hom}}(Y, Z) \cong Z \boxtimes GY.$$

- ② $Y \boxtimes -$ admits a left adjoint (internal coHom)

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If the category is abelian, $\otimes$ is right exact and $\boxtimes$ is left exact:

$$X \otimes Y \rightarrow Z \quad \text{fusion}$$

$$X \boxtimes Y \leftarrow Z \quad \text{fission?}$$

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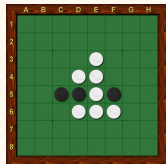
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Any statement about one monoidal product has a dual version using the other. Like *Othello/Reversi* stones that can be flipped to change colour or a photo negative.



Distributors in GV categories

Associators relate the two ways of combining two copies of either $\otimes$ or $\boxtimes$. Distributors relate the two ways of combining $\otimes$ with $\boxtimes$

$$\delta^l : X \otimes (Y \boxtimes Z) \rightarrow (X \otimes Y) \boxtimes Z, \quad \delta^r : (X \boxtimes Y) \otimes Z \rightarrow X \boxtimes (Y \otimes Z).$$

The distributors are generally **not** isomorphisms. They can even be 0-morphisms in certain examples.

Distributors in GV categories

Distributors must satisfy pentagonal constraints similar to those of associators.

$$\begin{array}{ccc} & (W \otimes X) \otimes (Y \otimes Z) & \\ \alpha^{\otimes} \swarrow & & \searrow \alpha^{\otimes} \\ ((W \otimes X) \otimes Y) \otimes Z & & W \otimes (X \otimes (Y \otimes Z)) \\ \alpha^{\otimes} \swarrow & & \searrow \alpha^{\otimes} \\ (W \otimes (X \otimes Y)) \otimes Z & \xrightarrow{\alpha^{\otimes}} & W \otimes ((X \otimes Y) \otimes Z) \end{array}$$

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Proposition [Cockett, Seely '97], [Fuchs, Schaumann, Schweigert, SW '24]

GV categories admit a canonical choice of distributors.

An analogue to rigidity in GV categories

Proposition [Fuchs, Schaumann, Schweigert, SW '24]

In every GV category $(\mathcal{C}, \otimes, K)$ there exist distinguished morphisms

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characterised by

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$$\mathrm{coev}_X \mapsto \mathrm{id}_{GX} \mapsto \mathrm{ev}_X$$

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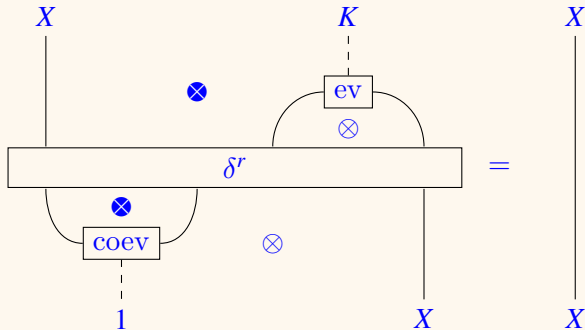
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Such that GV **zig-zag relations** hold:

$$\begin{array}{ccccccc} X & \xrightarrow{\mathrm{coev}_X \otimes \mathrm{id}} & (X \otimes GX) \otimes X & \xrightarrow{\delta^r} & X \otimes (GX \otimes X) & \xrightarrow{\mathrm{id} \otimes \mathrm{ev}_X} & X \\ & \searrow \mathrm{id}_X & & & & & \nearrow \\ & & & & & & \\ GX & \xrightarrow{\mathrm{id} \otimes \mathrm{coev}_X} & GX \otimes (X \otimes GX) & \xrightarrow{\delta^l} & (GX \otimes X) \otimes GX & \xrightarrow{\mathrm{ev}_X \otimes \mathrm{id}} & GX \\ & \nearrow \mathrm{id}_{GX} & & & & & \searrow \end{array}$$

GV rigidity in diagrams



Summary

- Grothendieck-Verdier duality is a natural duality structure for many algebraic representation categories, including vertex algebras.
- If **1** is dualising, GV duality provides natural candidates for evaluation and coevaluation. May be helpful for rigidity proofs.
- Many of the features needed for full/bulk/local conformal field theory can be naturally constructed in the GV context (not discussed here): GV module categories, GV Frobenius algebras, etc.
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Thank you for coming!